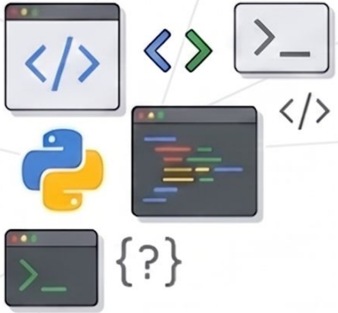


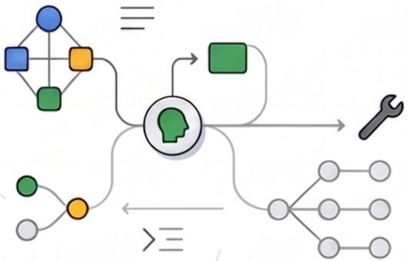
Welcome to

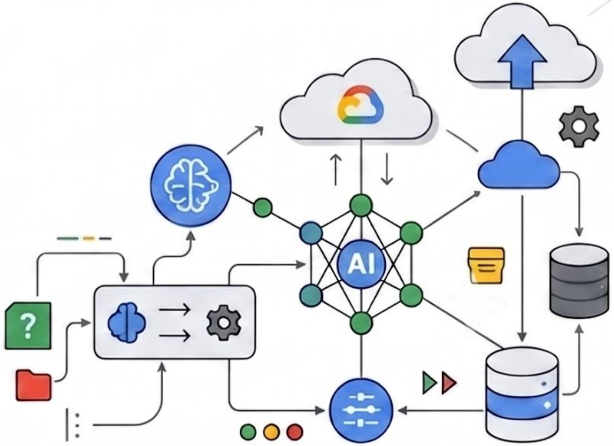
Python to Agentic AI Engineering

From Beginner to Production-Ready AI Agents with Google's AI Stack

Session 01

Course Introduction





Course Roadmap

- Module 1: Python Foundations
- Module 2: Python for AI Engineers
- Module 3: LLM Foundations
- Module 4: Building AI Applications
- Module 5: Agent Fundamentals
- Module 6: Google ADK
- Module 7: Multi-Agent Systems
- Module 8: Deployment with GCP

Google AI Studio

Gemini

Google Cloud

Agent Development Kit

Who this course is for?

- **Absolute beginners.**
- **Who want to learn how to build AI applications and AI agents using Python**.
- You do **not** need prior programming experience.
- If you can use a computer, browse the internet, and are willing to practice regularly, you can succeed in this course.

Pre-requisites:

- There are **NO** technical prerequisites.
- A laptop or desktop computer with internet connection.

How to get most of the Course?

- **Curiosity:** This is more important than prior knowledge.
- **Consistency:** Small, regular effort produces the best results.
- **A Growth Mindset:**
  - Willingness to Make Mistakes.
  - Learn by Building.
  - Focus on progress
- Don't just memorize, understand.
- Use AI as a Teacher, Do not just copy/paste.
- The ultimate goal is not mastering one tool—it's becoming someone who can adapt and keep learning.

What will you be able to do after this course?

- ✅ Understand Python code.
- ✅ Write Python programs confidently
- ✅ Use Generative AI models through APIs
- ✅ Build AI-powered applications
- ✅ Create AI agents using Google ADK
- ✅ Develop multi-agent systems
- ✅ Deploy AI agents on GCP Agent Engine
- ✅ Build portfolio projects that you can showcase to recruiters or use as the foundation for your own products and startups

Career Paths

Role	Skill level	Description
Python Developer	Beginner	Builds and maintains core backend software
AI Engineer	Intermediate	Build AI applications. Designs autonomous, task-executing AI systems.
Full Stack Developer	Advance	Build application (Frontend and Backend)
Data Engineer	Intermediate	Builds data pipeline and ETL processes
Startup Founder	Advanced	Ships user-ready, AI-powered applications

Course Structure

- Every session has:
    - Theory
    - Live Coding Demo
    - Practice Problems
    - Tasks to complete

After Every Sessions:

➔ Rewatch the video
➔ Revise the notes
➔ Complete practice exercises
➔ Experiment beyond exercises
➔ Ask questions if you are stuck

What you should Not do?

- Copy-Paste without understanding
- Depend only on AI
- Skip practice questions
- Wait until end of course

Course Resources:

- Git Repo: https://github.com/yazeedbashammakh/python-to-agentic-ai-course

How to Ask Questions

Good Question

I was trying to call Gemini API.

I expected:
A JSON response

I got:
Error XYZ

Here is my code:
...

Poor Question

Code not working.

Question Template

1. What were you trying to do?
2. What happened?
3. What did you expect?
4. Share error message.
5. Share code snippet.
6. Add proper Title, tags and select right category.

➔ This helps everyone solve issues faster.

Community Guidelines

Please remember:

1. Be respectful.
2. No question is too basic.
3. Help fellow learners when possible.
4. Search existing discussions before posting.
5. Keep discussions related to the course.

Instructors

Yazeed Bashamakh
Lead Instructor

Majed Khan
Associate Instructor

Shoeb Shaikh
Associate Instructor

Let’s Start

What is Programming?

- Programming is the process of giving instructions to a computer to perform tasks.
- Suppose you tell a friend:
  - Boil water.
  - Add tea leaves.
  - Add sugar.
  - Add milk.
  - Serve tea.
- Similar instruction to computer is called a **program**.
- We give instructions in programming language
  - Python
  - Java
  - Javascript
  - Php
  - C
  - C++

- Sample program:
  - Take 2 numbers as input
  - Sum them together
  - Give the sum

Automation

- Programming becomes powerful because it enables automation. Repetitive tasks can be defined once and executed multiple times.
- Before Automation
  - Teacher calculates marks manually.
  - Time consuming.
  - Mistakes might happen.
- After Automation
  - Computer calculates marks.
  - Fast, Consistent, less efforts.
- Examples:
  - ATM Machine
  - Sending Emails: Send 100 email automated.
- Automation helps us:
  - Save time.
  - Reduce errors.
  - Improve productivity.
  - Focus on more important work.

Python Language

- Python is a high-level, general-purpose programming language.

➔ Web development	➔ Data science	➔ AI
➔ Automation	➔ Big Data	➔ Cyber Security

Why Python?

- **Simple and Readable Syntax:** English-like language, Focus on logic not code syntax
- **Massive Ecosystem:** Many Libraries available like NumPy, Pandas, Matplotlib, Scikit-Learn etc.
- **Industry Adoption:** The Undisputed King of AI and Data Science

Getting Started with Python

1. Online IDEs:
   a. Google Colab - https://colab.research.google.com/
2. Python Installation - https://www.python.org/downloads/

IDE (Integrated Development Environment) Setup:

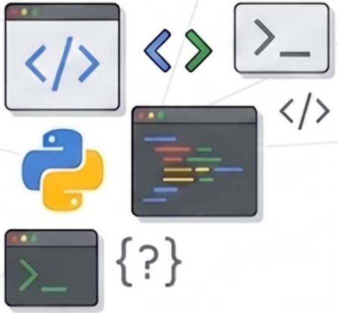
- VS Code - https://code.visualstudio.com/download → (Recommended)
- AntiGravity - https://antigravity.google/download → (AI powered)
- Cursor - https://cursor.com/download → (AI powered)

Local Python Installation (Optional)

1. Download python installer: https://www.python.org/downloads/
2. Run installer and follow the steps.
   a. Important: Check add python to PATH
3. Verify after installation:
   a. Check python version in cmd prompt or terminal:
      i. python3 --version

         or

      ii. python --version



Thank You